AUDITING GENERALIZATION CLAIMS FOR LLM AGENT HARNESSES: SEMANTIC BINDING AND MEASUREMENT VALIDITY

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ABSTRACT

In our deliberately tool-green semantic-binding benchmarks, a retrospective audit of 169 frozen agent trajectories with hash-verified verdict-to-submission pairing found that family-specific tool-success signals accepted every run, including all 82 semantically incorrect bindings. Typed provenance/authority oracles separate these semantically invalid successes from real ones by verifying whether submitted values occupy the roles the underlying evidence authorizes; across those trajectories the tool signal is constant, so all measured discrimination comes from the oracle. Measuring semantic binding this way, we then ask how far a harness intervention transfers, and introduce a *claim-scope framework* for harness interventions in LLM agents. Generalization here is not one axis but three—over task instances, over model backends, and over task families—and widening one is *incomparable* to widening another, not a further rung: crossing families implies nothing about crossing models, and conversely. We therefore index the *evidence* for a harness-effect claim by the set of (family, instance, model) configurations it occupies, ordered by inclusion, and require each dimension to be evidenced on its own. On a controlled tool-grounded *semantic-handoff* task, an instance-answer-independent "clarity bundle" (BundleS) is associated with zero observed axis-binding failures for Qwen3.7-Max where the ambiguous baseline fails two of three, and that direction reproduces on a pre-frozen held-out instance. At three repetitions per condition each cell is imprecise (Fisher $p = 0.40$; exact interval 29.2–100.0% vs. 0.8–90.6%), so we report an *observed and once-replicated* effect, not an established one. Widening the model dimension (DeepSeek-V4-Pro under exact counterbalancing) and the family dimension (a prospective twelve-instance STA panel; a second family at a non-discriminative ceiling) establishes nothing further: the STA panel gives $+12.5$ percentage points, moving over -12.5 to $+41.7$ under resampling of the panel, and its descriptive direction reversed relative to a three-instance pilot. The joint cell—different model *and* different family—was never measured, and we report it as untested rather than as a failed level or as covered by its projections. A second, orthogonal axis—a four-layer *Harness Effect Audit Protocol* (capability, sampling, execution, artifact integrity)—caught a concrete threat at every layer that would have changed the scientific conclusion. An external probe (Terminal-Bench 2.0→2.1, 26 repairs) finds the taxonomy relevant but incompletely covering (1 direct, 21 partial, 4 outside; sampling 0). Our instances test existence, recurrence, and transfer direction rather than estimate population-level pass rates.

1 INTRODUCTION

A commercial tool can sign off on a configuration whose values are bound to the wrong typed semantic roles: the tool is not malfunctioning—its answer is valid—but it answers a different question than the task poses, so tool success cannot decide task correctness. Our families are constructed so this is possible, and a retrospective audit shows real agents repeatedly enter that region (§3.1); typed provenance/authority oracles are what separate those cases, and are the grading substrate every number below already rested on. With a measurement that can see the failure, the question becomes how far an intervention that reduces it transfers.

When a harness intervention changes agent behavior, what can we claim? The question is usually answered along one implicit axis of "how general." But the unit a study actually realizes is a *configuration* $c = (f, i, m)$—a task family, a task instance belonging to it, and a model backend—so what indexes the *evidence* for a claim is the set of configurations at which it was measured, its **evidence support** $E \subseteq \mathcal{C}$, ordered by inclusion. The three familiar notions of generality are *projections* of that support:

- $s_I = \pi_I(E)$ **(instance scope):** the task instances a claim ranges over—$\{i_{\text{dev}}\} \subset \{i_{\text{dev}}, i_{\text{held}}\} \subset$ a declared frozen panel $\subset$ the generator's instance population.
- $s_M = \pi_M(E)$ **(model scope):** the backends—{Qwen3.7-Max} $\subset$ {Qwen3.7-Max, DeepSeek-V4-Pro} $\subset$ a model population.
- $s_F = \pi_F(E)$ **(family scope):** the task families—{workflow} $\subset$ {workflow, STA}, and so on up to a family population.

We index by the support and not by the triple (s_I, s_M, s_F), for two reasons. First, **instances are family-specific**: an STA instance is not a workflow instance evaluated elsewhere, so a family-widening probe necessarily brings its own new instance identities and s_I and s_F are not free coordinates of a product. Second, and consequently, $\pi_F(E) \times \pi_I(E) \times \pi_M(E) \neq E$ in general—a support is typically *sparse* in the product of its own projections—so having widened every projection somewhere is not the same as having measured every configuration. That gap is precisely where this study's central omission lives, and a triple of marginals cannot express it. Finally, E is what *audits* a claim, not what the claim asserts: a claim may name a wider **target support** T, and $E = T$ only when nothing is generalized. The lattice orders E; the standard of §3.1 governs which relations between E and T a design licenses.

Enlarging the support requires evidence *inside* the enlargement, and inclusion is a **partial order, not a chain**. The two ways a harness result is usually generalized—"it also works on another model" and "it also works on another task family"—add **incomparable** configurations: neither support contains the other, so neither is a prerequisite for the other, and an intervention can cross every family for one model while failing on a second model. Arranging them as successive rungs of a ladder would make a claim look better-indexed than its evidence, and would license the invalid inference that a cross-family result presupposes a cross-model one. Failure at a larger support does *not* refute a claim at a smaller one; it bounds the claim's scope.

Our frozen evidence realizes four of the five named scope regions below, and does so across five distinct supports (Figure 1): **S0** (local—Qwen3.7-Max, workflow family, development instance); **S1** (=S0 plus a pre-frozen held-out instance, widening s_I); **S2-M** (=S0 plus DeepSeek-V4-Pro, widening s_M); **S2-F**, which is a *class* of two incomparable family-widening supports (STA and SPICE) rather than one; and **S3**, any support holding a configuration whose family is not the workflow family *and* whose model is not Qwen3.7-Max. S2-M and S2-F are siblings, not consecutive levels. **S3 contains no configuration we measured.** We report it as untested rather than as a failed level—and, because the projections do not reconstitute the support, not as covered by the fact that s_M and s_F have each been widened somewhere.

Because each S0/S1 cell rests on one instance at k=3, we call that result *observed and once-replicated* and reserve "established" for the standard of §3.1; Table 2 gives the numbers.

A harness-effect claim has two independent questions: (A) *what scope of generalization does the evidence support?* and (B) *was the measurement itself valid?* The scope lattice addresses (A); a four-layer audit protocol (capability, sampling, execution, artifact integrity) addresses (B). Figure 1 places our concrete evidence and incidents into this 2D framework.

Contributions. (1) **Semantically attested evaluation.** Typed provenance/authority oracles, instantiated across three executable EDA task families, separate professional-tool success from correct semantic role binding; a retrospective, hash-paired audit of frozen trajectories shows the family-specific tool-success signal was *constant* over all 169 retained runs and accepted all 82 observed semantic misbindings (§3.1). The tasks are deliberately constructed so wrong bindings can remain tool-green, so this quantifies real-agent occupancy of that failure region rather than estimating its prevalence in agent benchmarks generally. (2) The semantic-binding failure itself: two subtypes we never collapse—axis-binding, and role-conditioned value selection—exhibited as real, tool-green agent trajectories. (3) A claim-scope framework indexing the *evidence* for a harness-effect claim by the configurations it occupies rather than by a single ladder, making instance, model and family generalization distinct and partially ordered and making explicit what a study leaves unmeasured, with a qualification standard applied symmetrically to our own positive and negative results; applied to the semantic-binding study it yields an observed, once-replicated local effect with no established transfer along either wider dimension, including a prospective 12-instance expansion whose descriptive direction reversed relative to a 3-instance pilot. (4) A measurement-validity framework in which a verdict must be bound to the artifact it claims to judge and an oracle inherits the custody of its reference standard: capability, sampling, execution and artifact failures each materially changed scientific interpretation, plus external taxonomy evidence from Terminal-Bench repairs.

2 Related Work and Problem Formulation

Harnesses and harness effects. A *harness* is the task-level information structure visible to the agent (prompt, visible files, disclosure bundle, public tool feedback, action surface). A *harness effect* is a change in semantic-binding correctness attributable to harness information content, holding the hidden truth, grader, and tool fixed.

Problem formulation. Each task is a *semantic handoff*: bind a tuple to canonical typed roles from role-misleading evidence. The workflow family binds PVT descriptors to typed axes; Family A (STA/PrimeTime) binds (intent_class, target_partition, check_mode) from an authority provenance DAG; Family B (SPICE/HSPICE) binds (corner, load_condition, metric) from a request–authority join.

TAB vs. semantic binding. Task alignment (Mavali et al., 2026) asks *which* environmental evidence an agent should follow; semantic binding asks *which semantic role* selected evidence is permitted to fill. Our failures can occur after relevant evidence has been retrieved and can remain tool-green—the agent finds the right artifact but binds its value to the wrong typed axis.

Positioning. Yao et al. (2026) broadly characterize model$\times$harness interactions; Wang et al. (2026a) critique harness-leaderboard conflation; Ursekar et al. (2026) optimize harnesses. We do *not* claim priority over the general observation that a reported success can be unearned; four recent works bound our claim, and what remains ours is narrower. Zhu et al. (2025) and Shao et al. (2026) set benchmark-construction and protocol-validity standards, the latter formalizing whether the target capability stays necessary for success and quantifying shortcut-driven score inflation—but our agents do not bypass the protocol: they complete it, a real commercial tool accepts the artifact, and the values still occupy the wrong roles. Gao & Zhou (2026) asks whether stored evidence supports a claimed outcome at all, labelling unsupported runs Unknown and widening score bounds accordingly; we address the complementary case in which the evidence is complete and the tool genuinely accepts the final artifact. Fan et al. (2026) is closest conceptually—semantic roles for tool arguments, provenance tracking, role-specific authority contracts—but targets runtime security enforcement rather than benchmark measurement, so semantic roles plus provenance is not ours to claim as new. Luo & Peng (2026) audits whether a correct outcome drew on permitted information; our object is value-to-role authority binding, not target-answer exposure. Ma et al. (2026) report cross-benchmark harness gains, which we use as evidence that some interventions do transfer—so transfer must be demonstrated rather than assumed. Three further concurrent works and the transfer-dimension literature are positioned in Appendix J; against all of them, what we add is the claim-scope lattice, a controlled semantic-binding mechanism, and an audit of which measurement failures changed the scientific conclusion.

3 Methodology: The 2D Framework

The claim-scope lattice (first axis). The support E and its partial order are defined in §1. Two consequences govern the studies below: a study may legitimately measure S2-F without S2-M, but may present neither as subsuming the other nor treat the unmeasured S3 as a verdict reached; and because instances are family-specific, a cross-family verdict is never a pure single-projection move—a property of the design space rather than a confound, which we flag where it matters (§3.1).

The measurement-validity axis (second axis). Any measurement has four distinct validity dimensions: (1) capability validity (does the task measure the intended capability, including tool-green semantic alternatives?); (2) sampling validity (repeated trajectories, counterbalancing, membership arbitration); (3) execution validity (transport, terminal-vs-recovered, tool health, protocol completion); (4) artifact integrity (action-surface isolation, canonical-source integrity, custody). These are complementary audit dimensions, *not* independent factors and *not* an ordering: they interact, and a fault in one routinely presents as a symptom of another—our canonical-source mutation (Layer 4) surfaced as an apparent tool-health failure (Layer 3), which is exactly why the protocol records which layer a fault actually belonged to rather than which layer raised the alarm.

Figure 1: The 2D framework: claim scope (columns) $\times$ measurement validity (rows). Columns are supports in the claim lattice, *not* a progression: S1, S2-M and S2-F each enlarge S0 along a different projection, so S2-M and S2-F are incomparable siblings. S3 enlarges both s_M and s_F and is the region this study leaves empty—a state a ladder cannot distinguish from a failed test, and one that widening each projection separately does not fill.

	S0 local	**S1** $+s_I$ held-out	**S2-M** $+s_M$	**S2-F** $+s_F$	**S3** $+s_M+s_F$
Cap.	tool-green wrong binding; typed oracle	BundleS 3/3 (direction replicated)	DeepSeek 3/4=3/4	STA n=12 not established; SPICE ceiling	*not measured*
Samp.	position-balanced blocking; non-det.	pre-frozen held-out inst.	counterbalanced block	prospective 12-inst. schedule	*not measured*
Exec.	SSE streaming (censoring caught)	recovered degradation kept	—	SPICE action-surface repair	*not measured*
Art.	canonical-tree guard	custody hashes	—	—	*not measured*

3.1 Claim qualification: what "established" is allowed to mean

Indexing a claim by scope still leaves open how strong the evidence at a given support must be before a word like "established" is used. We state the standard in advance and—this is the part a claim-auditing paper most owes its reader—apply it symmetrically, at the same bar for our positive result as for our negative ones. Four tiers, defined in full in Appendix E: **observed** (a descriptive separation at that support, no target beyond E); **replicated** (the direction reappears at *newly frozen* configurations enlarging the projection claimed—re-running the same configurations later is repeated measurement, never replication); **estimated on a fixed panel** ($T = E$, accompanied by a sensitivity band rather than a confidence statement); and **generalized** (the one tier asserting $T \supsetneq E$, requiring a distribution the design instantiates).

Under this standard the BundleS result is *observed* at S0 and *replicated* at S1—a replication in s_I, since S1 adds a new instance rather than re-running the old one; it is *estimated* nowhere in the workflow family, since each cell is one instance at k=3 (intervals 29.2–100.0% against 0.8–90.6%, two-sided Fisher $p = 0.40$), and it is *generalized* nowhere at all. The S2-F panel is *estimated* and not generalized. Table 1 gives the general mapping; every verdict in this paper is phrased to sit inside its middle column.

Table 1: Claim qualification: what a given evidence shape does and does not license. The right column collects wordings that recur in harness-effect reporting but that the evidence does not support.

Evidence shape	Licensed	Not licensed
One instance, repeated trajectories	effect observed on this instance	improves the task family
One pre-frozen held-out instance	direction replicated once	within-family generalization established
Frozen multi-instance panel	realized-panel average with a composition-sensitivity band	population transfer
Same configurations, new run window	stability of the measurement	replication at any wider scope
One alternate model backend	transfers, or fails to, on that backend	model-general effect
A family at ceiling	uninformative for this contrast	no transfer
Retrospective external coding	taxonomy relevance	taxonomy validity

Construct: what semantic correctness means. Semantic correctness is *not* inferred from tool success, a hidden numeric answer, or the artifact score. It is decided solely by whether the submitted tuple matches the binding attested by the independent provenance/authority oracle. *Worked example (Family A):* visible evidence attests intent=*functional_close*, partition=*core*, check_mode=*setup*; the agent submits (functional_close, core, *both*)—PrimeTime returns green, the value is type-valid, but the coverage authority attests *setup*, not *both*; the oracle rejects it as a role-conditioned value-selection failure. This construct is instantiated independently in the workflow family (PVT axes), Family A (intent/partition/check_mode), and Family B (corner/load/metric), each with its own grader. We do *not* claim human validation (Study B was preregistered but unexecuted).

A worked instance (workflow family, ambiguous variant). The agent receives an out-of-date timing sign-off handoff and must repair `flow_config.json` to the unique package (netlist, clock, scenario, corner), then regenerate a two-stage evidence chain. The difficulty is deliberately *not* retrieval: the reports label their axis fields `op_point` and `mode`, and this variant ships no glossary or axis declaration, so which logical axis each label denotes must itself be recovered from the intersection of the sources. Five constraints are recoverable from the specification—C1 netlist family, C2 clock identity (the only clock with non-zero intended-clock coverage), C3 scenario-typed, C4 corner-typed, C5 the sign-off pair. Exhaustive enumeration over the declared domains yields 294 candidate assignments, of which **exactly one** satisfies C1–C5: (`netlist_v2.v`, `clk_main`, `slow`, `func`).

Four evidence artifacts are shipped, each *pairwise plausible* and each violating a different constraint (Table 4, Appendix A). Source A is the critical one: it carries the right netlist *and* the right clock, and **PrimeTime signs it off green**—but its two role fields are swapped, so a corner value occupies the scenario role and a scenario value the corner role. Every shipped source signs off green, so no tool signal separates A from the truth—only the typed oracle does. This is what *tool-green* means throughout the paper, and it is why a benchmark that scores this task by tool exit status would score a wrong binding as a pass.

How the oracle adjudicates. The grader never consults tool exit status when deciding semantic correctness. It applies typed-membership predicates to the submitted package: the scenario field must belong to the scenario axis and the corner field to the corner axis, neither may be a PVT descriptor (a descriptor denotes a scenario–corner *pair* and is never a value of either axis), and the clock must match by exact identity, so a generic alias is rejected. These predicates are folded into the master evidence gate, so a sign-off-green but mis-typed package cannot pass. A failure is classified into exactly one of two subtypes we never collapse: an **axis-binding failure**, where a value occupies the wrong typed axis (e.g. `func`/`slow`), and a **role-conditioned value-selection failure**, where the value is axis- and type-valid but the wrong member fills the role (e.g. `typ`/`func`). Uniqueness is verified by exhaustive enumeration before an instance is admitted, so "correct" is fixed by the instance's constraint system, not by grader judgement.

What the oracle discriminates that the tool does not. The oracle's necessity is usually argued from construction—a wrong binding *can* stay tool-green. A **retrospective, post-freeze audit** (specified after the program closed, not preregistered, adding no episode, model call or tool run) measures the consequence. Pairing each episode's semantic verdict with its family's own pre-existing tool-success field, and excluding rather than imputing 33 of 202 considered trajectories, the tool signal equals *accept* for all 169 retained runs across five strata—so all 82 semantic errors are accepted. Against the target variable it is a constant with **zero empirical discrimination** between the 87 correct and 82 incorrect submissions, and all discrimination the study reports comes from the oracle. This is not a population false-accept rate: strata are never pooled, and the families are built so a wrong binding survives the tool. Nor is it two findings—with the signal constant, $\Delta \equiv 1 - S_{\text{semantic}}$. What construction alone does *not* give is behavioral: a region reachable in principle was occupied by real agents 82 times. SPICE phase-5D is the negative control (18/18 correct, oracle and tool agreeing), so the oracle does not manufacture disagreement. Appendix F gives the per-stratum table, exclusions, and pairing sensitivity.

BundleS (canonical labels, value-domain definitions, glossary, procedural contract; component C6 withheld; no golden values); **TypedContract** (same information as BundleS as a JSON Schema). Hidden truth and grader are identical across conditions. We call BundleS **instance-answer-independent**

rather than "non-answer-bearing": it withholds the instance's target tuple but deliberately discloses *task-level role semantics*—which axis a label denotes and what values each axis admits—so calling it answer-free would understate it. The same schema serves every instance, so it contains no explicitly instance-specific assignment field; whether schema and generator structure *jointly* permit recovering an assignment is a separate question, and §7 gives the leakage check that would settle it.

Design and models. Seeded exact-counterbalanced blocked randomization, frozen before any paid call. Instance is the primary unit; repetitions nested. We evaluate Qwen3.7-Max and DeepSeek-V4-Pro; these are not representative of the frontier-model population. The design is diagnostic, not population-estimating: it tests existence, recurrence, ceiling/null behavior, and transfer direction; it cannot resolve small effect sizes.

4 Study I — Local Effect and Held-Out Replication (RQ1; S0–S1)

On the workflow controlled pair (identical truth and grader; ambiguous vs. clear handoff), BundleS was associated with *zero observed* axis-binding failures for Qwen3.7-Max where the ambiguous baseline failed two of three: Base 1/3 $\rightarrow$ BundleS 3/3. On a *pre-frozen held-out* instance the direction reproduces: Base 1/3 $\rightarrow$ BundleS 3/3, again with no axis-binding failure under BundleS. Each is one instance at k=3 and therefore individually imprecise—exact intervals 0.8–90.6% for Base against 29.2–100.0% for BundleS, two-sided Fisher $p = 0.40$ at S0 and $p = 0.40$ at S1—so the evidential weight lies not in either cell's separation but in the direction reappearing at an independently frozen instance. The subset producing it withholds C6, so the effect is not attributable to explicit C6 disclosure of the assignment. The failure taxonomy partitions the workflow ledger (70 episodes) into 41 correct and two subtypes we never collapse—24 *axis-binding* and 5 *role-conditioned value-selection*—all tool-green, rejected only by the typed oracle.

A sequential decomposition tested whether a stable minimal component underlies BundleS. None was isolated: C1 alone, C2-only, C4-only, and a C24 bridge all failed their thresholds. We therefore describe this as an *exploratory component localization that did not converge*, not as a demonstration that no component mechanism exists—failure to identify is not absence, and at k=4 per condition the design has little power to distinguish the two. Table 5 (Appendix B) gives the full per-condition ledger behind both statements.

Three things in that ledger are worth reading directly. First, the *full* bundle—which does include C6, the component that discloses the instance's assignment—moves both models from failure to 3/3; the scope limitation we report at S2-M is about the instance-answer-independent subset, not about the phenomenon. Second, the axis column is where the mechanism lives, but it does not localize to one ingredient: every condition that suppresses axis errors carries value-domain information, yet C2-only and C4-only carry it too and still retain three axis failures each, so neither is sufficient alone—and the C1-alone row refutes the pre-declared hypothesis that canonical labels alone eliminate axis errors. Third, three condition–model pairs were measured in more than one run window and every one disagrees with itself: C24 gives 3/4 with no axis error then 2/4 with one, Base(0009)/DeepSeek gives 0/3, 2/3 and 3/4, and BundleS/DeepSeek moves from 1/3 to 3/4. That within-condition spread *across* windows, visible only because the ledger does not deduplicate repeats, is the direct evidence that a component result at k=4 is not stable enough to name a mechanism. One caveat: those windows span 22 days and our records pin the *requested* alias, not a provider-resolved snapshot, so stochasticity and backend drift cannot be separated (§6). Both readings support the inference we draw, but only the first is a statement about the model.

Verdict at S0 and S1: *observed*, and *replicated* once (§3.1); not *established*, and nowhere *generalized*. The separation is descriptive within each cell, and the replication rests on one independently frozen instance, so neither licenses a family-level treatment-effect claim; we make no universal improvement claim. A negative result at S2-M or S2-F does not refute this—it bounds the coordinate along which the effect has been shown, which is the entire purpose of indexing the claim.

5 STUDY II — WIDENING THE MODEL AND FAMILY COORDINATES (RQ2; S2-M, S2-F)

Does the S0–S1 effect survive widening s_M or s_F? Table 2 is the main result. The two questions are asked separately because they are separate coordinates: a verdict at S2-M constrains nothing at S2-F.

Table 2: Main result matrix (semantic-binding correctness; instance is the unit). Workflow/SPICE entries are correct of k stochastic repetitions; STA entries are the mean correct-rate over 12 instances. Verdicts use the qualification standard of §3.1 and Table 1, so "observed" and "replicated" are deliberately weaker words than "established." **S2-F is a class of supports, not one:** STA and SPICE are two separately realized family-widening probes, with disjoint instance sets and family projections $\{$workflow, STA$\}$ and $\{$workflow, SPICE$\}$, neither of which contains the other. The last row is the joint cell this study does not populate; it is listed so that an unmeasured scope is visible in the main result rather than inferable only from its absence.

Family / model	Base	BundleS	TypedContract	#inst.	verdict
Workflow / Qwen3.7-Max (S0)	1/3	3/3	—	1	observed
Workflow / Qwen3.7-Max (S1, held-out)	1/3	3/3	—	1	replicated once
Workflow / DeepSeek-V4-Pro (S2-M)	3/4	3/4	—	1	not established
STA / Qwen3.7-Max (S2-F, prospective)	**0.208**	**0.333**	**0.458**	**12**	**estimated; band spans 0**
SPICE / Qwen3.7-Max (S2-F)	1.00	1.00	1.00	3	ceiling; uninformative
joint model×family (S3)	—	—	—	0	*not measured*

S2-M (widening the model coordinate). Under exact counterbalancing, BundleS is not established for DeepSeek-V4-Pro (3/4 = 3/4; exact interval 19.4–99.4% in both arms, two-sided Fisher $p = 1.00$). This does not refute the S0–S1 result for Qwen3.7-Max; it bounds the model coordinate. Neither does it establish absence: at k=4 that interval is wide enough to contain effects of practical size, so "not established" here means the design could not resolve the question, not that the answer is no.

S2-F (widening the family coordinate, STA). Across 12 prospectively frozen instances the realized panel difference is $+12.5$ percentage points (BundleS $-$ Base), and resampling the twelve instances moves that estimate over -12.5 to $+41.7$. We report this as an *instance-resampling sensitivity band* and not a confidence interval—the declared estimand is the realized panel, whose composition carries no sampling uncertainty, so the band measures dependence on the panel's empirical composition and straddles zero as well as effects of practical size in both directions (Appendix C states the limits of that reading). We lead with the point estimate and its sensitivity because the preregistered statistic—exact sign-test $p = 1.0$ on 3 improving, 2 declining and 7 tied instances, permutation sensitivity $p = 0.31$—is easily misread as evidence *for* a zero effect, which it is not; the frozen analysis is unchanged, remains the sole basis for the confirmatory verdict, and agrees that nothing is established. *The three-instance pilot (-16.7pp) and the prospectively frozen twelve-instance panel ($+12.5$pp) reversed direction. We read that as evidence that very small task panels are unreliable for directional harness claims, not as evidence that BundleS transfers; small-panel instability of directional benchmark claims has been independently documented for agent-safety benchmarks (Wang et al., 2026b).* TypedContract is descriptively highest (0.458) but no preregistered TypedContract advantage was established; this is hypothesis-generating only. The pilot (n=3) is not pooled with the prospective n=12; per-instance values for both are in Appendix C so the reversal is checkable rather than asserted.

S2-F (SPICE). Base = BundleS = TypedContract = 1.00—a non-discriminative ceiling. This is a measurement limitation, not evidence against transfer.

Three outcomes, kept distinct. Widening s_F yields three things we never pool: *no established effect on a discriminating family* (STA, n=12), unable to rule out a small effect; a *non-discriminative ceiling* (SPICE); and a *direction reversal under prospective expansion*. None is evidence about S2-M, and none populates S3.

6 STUDY III — MEASUREMENT VALIDITY AND EXTERNAL SCOPE PROBE (RQ3)

Each headline conclusion could have been manufactured or masked by a measurement artifact. The four-layer protocol is instantiated and stress-tested; Table 3 lists incidents where a control *materially changed the interpretation*.

Table 3: Audit-protocol incidents: each control prevented or corrected an invalid scientific interpretation.

Threat	Control	Conclusion that would have been wrong
Tool-success proxy (Cap)	Typed provenance/authority oracle	Tool-green wrong bindings scored as correct
Non-streaming censoring (Exec)	SSE + terminal-transport arbiter	Qwen3.7-Max falsely recorded as failing
Position confounding (Samp)	Exact position-balanced blocking	Base/BundleS order leaked into contrast
Recovered degradation (Exec)	Terminal-valid vs recovered split	Hard-but-recovered solve discarded
PT corrupted measurement (Exec)	Tool-health sentinel + bookends	Tool hiccup read as model failure
SPICE action-surface false positive (Art)	Immutable core + forensic audit	12 SPICE episodes spuriously zeroed; semantic-binding ceiling obscured
Canonical-source mutation (Art)	Exact-commit worktree + hash guard	Golden rewrite read as tool outage

Why the controls materially affected interpretation. For three of these rows the final column is not hypothetical but the reading the uncorrected data would in fact have produced: a headline inversion in each case, not a detail.

A verdict is only about the artifact it can be shown to describe. Comparing a tool verdict with a semantic one presupposes both describe the same submission, which for the workflow family does not hold automatically: the agent runs its evidence chain mid-episode and may edit the configuration afterwards. Because the stage-2 record stores the hash of the configuration it consumed, hash-level attestation excluded 9 of 58 historical workflow runs whose tool verdict did not describe the final submitted artifact—tuple equality would have caught two of the nine. These are not nine mis-scored episodes; the frozen grader had already marked the broken stage chain. They show a *post hoc* paired analysis must re-establish verdict-to-artifact binding rather than assume it. STA and SPICE satisfy it by construction.

A monitor is only as trustworthy as the custody of its reference standard. Our tool-health monitor compares a reference artifact against a frozen fingerprint and reports the environment unhealthy on mismatch. During development it fired for roughly a day and the outage was attributed to the remote tool server. The monitor's logic was correct and its mismatch real; the attribution was wrong, because the artifact it measured against had itself been rewritten—not by the remote server, but by a component of our own test harness writing into the canonical tree. A monitor validating a system against a reference standard therefore inherits the custody of that standard: if the standard can be mutated by the harness, the monitor faithfully reports a genuine mismatch while misattributing the fault. We treat this as a fourth Layer-4 requirement: *the reference standards of health checks must be isolated from the harness whose health they certify.* The affected artifact was in the development workspace, already treated as non-authoritative; no reported measurement derives from it.

Model custody: a fifth Layer-4 requirement, stated as our own gap. The custody argument generalizes past files: a measurement is reproducible only if the *system under test* is identified as precisely as the artifacts are, and against a commercial API that system is named by a provider-controlled alias. Episodes span 22 days (2026-07-12 to 2026-08-03), yet a provider-resolved snapshot, response identifier or system fingerprint was retained for 0 of the 70 ledger episodes—the per-episode transport record has no such field, so the gap is total rather than partial—and only 66 retain even the *requested* alias and run date. Where an alias can be repointed between windows, two measurements of the “same” condition are not guaranteed to measure the same system. We therefore add a fifth Layer-4 requirement—*record the provider-resolved model identity per episode, not the requested alias*—and state it as a limitation we incurred rather than a control we exercised: for the across-window spread of §4 we cannot separate stochasticity from backend drift.

External taxonomy scope probe (Terminal-Bench 2.0→2.1). As an external scope probe (not validation), we coded the 26 officially documented repairs under the four-layer-plus-Other rubric, *frozen before coding*: **1 direct, 21 partial, 4 outside** (of 26); layers: capability 17, sampling 0, execution 10, artifact 1; 6 multi-layer. We interpret this as evidence that the taxonomy has *external relevance but incomplete coverage*. This is static retrospective taxonomy application; it does *not* validate runtime portability and does *not* validate every audit layer (the sampling layer received no external support). The full 26-task coding is in the Appendix.

7 Limitations and Discussion

Controlled executable tasks, not sampled production incidents. EDA-heavy environments—all results are scoped to EDA. *Small instance counts*—S0, S1 and S2-M each rest on a single-instance cell at k=3–4, with S1 using a distinct pre-frozen held-out instance; the design is diagnostic, and no cell reaches the *generalized* tier of §3.1. *Construct validity*—correctness is decided by executable provenance/authority oracles, internally consistent and instance-unique but *not* shown to agree with expert human judgement. The preregistered blinded human study (Study B) is unexecuted, so the construct we validate is a formal one, and we cannot exclude that some Base failures reflect underspecification a domain expert would also judge underspecified. *Model scope*—two backends, identified only by provider alias (§6); a second backend probes the model coordinate and does not yield a model-general result. *Family scope*—the three families are independently constructed but all authored by us and all within EDA, so S2-F is better read as cross-generator transfer inside one designed universe than as domain transfer. *Audit protocol*—the four layers were *derived from* the incidents we encountered, so Table 3 is framework-construction evidence and the Terminal-Bench coding is its first external evaluation, coded by the authors with no second independent coder and no agreement statistic. *No mechanism isolated*—an exploratory localization that did not converge, not a demonstration that none exists.

What would change these conclusions. Because the framework’s whole purpose is to make a claim’s scope contestable, we state in advance what evidence would move each verdict; none of it requires trusting our interpretation.

The seven conditions—one per verdict—are listed in Appendix D.

Discussion. Two results are not contestable by re-analysis at all, being properties of the measurement: the transport-censoring incident and the action-surface false positive would have produced confident wrong numbers in any analysis of the uncorrected logs. More broadly, because inclusion is partial and a support is sparse in its own projections, the lattice exposes three errors a ladder invites—reading one larger sibling as evidence for another, reading an unmeasured support as a failed one, and reading separately widened projections as joint coverage. Some harness interventions do transfer (Ma et al., 2026); transfer is therefore demonstrated at each coordinate, never inherited.

8 Conclusion

Professional tool success did not discriminate semantic correctness anywhere in our frozen trajectories: the signal was constant across all 169 retained runs, accepting all 82 observed misbindings, so typed provenance/authority oracles carried the entire measurement. Measuring on that basis, we indexed the evidence supporting a claim by its measured configuration support—making instance, model and family generalization partially ordered projections rather than rungs of a ladder, with an explicit standard for when a claim may be called established. Applied to our own results it yields an effect observed at S0, replicated once at S1, unestablished at S2-M and S2-F, and unmeasured at S3; a prospective 12-instance expansion reversed the descriptive direction without establishing anything. Each audit layer changed a conclusion, and our own model custody proved weakest.

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Table 4: The four shipped evidence sources of the worked instance. Each is pairwise plausible and each signs off green under PrimeTime, yet each violates a different typed constraint: A and D swap the two role fields, B is out of the netlist family, C puts a PVT descriptor in the corner role and a generic alias in the clock role. The hidden role mapping (`op_point`→scenario, `mode`→corner) is not disclosed in this variant.

source	netlist	clock	op_point	mode	violates	tool
A role swap	`v2` ✓	`clk_main` ✓	`func`	`typ`	C3, C4	green
B stale	`v1` ✗	`clk_main` ✓	`slow`	`func`	C1	green
C PVT label	`v2` ✓	`clk` ✗	`slow`	`slow_1.0V_125C`	C2, C4	green
D JSON chain	`v2` ✓	`clk_main` ✓	`func`	`typ`	C3, C4	green
unique truth	`v2`	`clk_main`	`slow`	`func`	—	green

Table 5: **Study I episode ledger**, one row per (stage, condition, model): typed-binding correct$/k$, then the two failure subtypes. *Inclusion rule:* the 58 program-primary episodes declared by the frozen ablation-phase manifests—omitting only what those manifests mark excluded, invalid or aborted (3, 0 and 1)—plus the 12 earlier controlled-pair episodes, giving 70; this descriptive ledger is deliberately broader than the 58-episode program-primary accounting of the reproducibility statement. A condition measured in two run windows occupies two rows and is never deduplicated, which is why Base(0009)/DeepSeek appears three times and C24 twice.

stage	condition (task)	model	correct$/k$	axis	value
controlled pair	Base(0009)	DeepSeek	0/3	3	0
	Full bundle(0010)	DeepSeek	3/3	0	0
	Base(0009)	Qwen	1/3	2	0
	Full bundle(0010)	Qwen	3/3	0	0
C6 ablation	V1 (0013)	Qwen	3/3	0	0
	V9 (0014)	Qwen	3/3	0	0
screening	BundleD (0016)	Qwen	1/3	2	0
	BundleS (0015)	Qwen	3/3	0	0
held-out	Base(0011)	Qwen	1/3	1	1
	Held(0017)	Qwen	3/3	0	0
cross-model 1	Base(0009)	DeepSeek	2/3	1	0
	BundleS (0015)	DeepSeek	1/3	2	0
cross-model 1C	Base(0009)	DeepSeek	3/4	1	0
	BundleS (0015)	DeepSeek	3/4	1	0
decomp. 1	Contract (0019)	Qwen	1/3	2	0
	Schema (0018)	Qwen	2/3	0	1
decomp. 2	C1 (0020)	Qwen	2/4	2	0
	C24 (0021)	Qwen	3/4	0	1
decomp. 3	C2 (0022)	Qwen	1/4	3	0
	C4 (0023)	Qwen	0/4	3	1
C24 bridge	C24 (0021)	Qwen	2/4	1	1
total			**41/70**	**24**	**5**

Base = ambiguous handoff; Full bundle = C1–C7 incl. C6; V1/V9 = C6 removed from the clear instance / added to the ambiguous one; BundleS = C1+C2+C4+C7 (no C6); BundleD = C3+C5 (decoy disambiguation); Held = held-out instance + BundleS; Schema = C1+C2+C4; Contract = C7 alone; C1 = axis schema; C2 = PVT definitions; C4 = glossary; C24 = C2+C4.

A Worked instance — shipped evidence sources

B Study I — per-instance ledger and minimal-component ablation

Every cell is re-derived from the preserved per-episode records (`flow_config.submitted.json`, `result.json`, sanitized agent log) by a committed script and is typeset directly from that script's output, so the table cannot drift from the records by transcription. The script also enforces the inclusion rule mechanically—it asserts each stage's episode count against the frozen program manifest and aborts on mismatch—because an earlier aggregation of these same records keyed cells by (condition, model) in a dictionary and so silently dropped repeat measurements of a condition; that is a failure mode a stated rule alone does not prevent. Transport is SSE streaming throughout except the DeepSeek controlled-pair rows, which are frozen non-streaming and reported as such. Program-wide: 70 episodes (58 program-primary plus 12 controlled-pair), 41 correct, 24 axis-binding, 5 role-conditioned value-selection. Minimal-component ablation (exploratory): no stable sub-component isolated—C1 alone, C2-only, C4-only and the C24 bridge all failed their pre-declared thresholds.

Per-cell uncertainty and a pooled figure we do not use. Each S0/S1/S2-M cell is one instance at k=3–4, so its exact interval is wide: BundleS 29.2–100.0% against Base 0.8–90.6% at S0 and S1, and 19.4–99.4% in both arms at S2-M. Two-sided Fisher gives $p = 0.40$ at S0, $p = 0.40$ at S1, and $p = 1.00$ at S2-M. For transparency: pooling the S0 and S1 cells into a single 2×2 table gives $p = 0.06$. We do *not* report that as a result—the design declares the task instance as the unit of analysis, and pooling two instances assumes an exchangeability the design does not establish—but we state it so that a reader can see the pooled figure was computed and set aside deliberately rather than overlooked.

C STUDY II — PROSPECTIVE STA DETAIL

Table 6: Prospective STA per-instance outcomes (n=12; Y=correct, n=wrong). These twelve instances are the realized panel that the S2-F estimate refers to; the sensitivity band of the main text asks how far that estimate moves when this membership is resampled.

	2-rep (rate)			diff		
inst.	Base	BundleS	TC	S−B	TC−B	TC−S
0004	nY(.5)	nn(0)	nn(0)	−.5	−.5	.0
0005	nn(0)	nn(0)	nn(0)	.0	.0	.0
0006	nn(0)	nn(0)	nn(0)	.0	.0	.0
0007	Yn(.5)	nn(0)	nn(0)	−.5	−.5	.0
0008	nn(0)	nn(0)	nn(0)	.0	.0	.0
0009	nn(0)	nn(0)	nn(0)	.0	.0	.0
0010	nY(.5)	YY(1)	YY(1)	.5	.5	.0
0011	nn(0)	YY(1)	YY(1)	1.0	1.0	.0
0012	nn(0)	YY(1)	YY(1)	1.0	1.0	.0
0013	YY(1)	YY(1)	YY(1)	.0	.0	.0
0014	nn(0)	nn(0)	Yn(.5)	.0	.5	.5
0015	nn(0)	nn(0)	YY(1)	.0	1.0	1.0

The historical pilot (n=3) is *not* pooled with the prospective panel at any point. Because the paper rests a methodological claim on the prospective expansion having reversed the pilot's descriptive direction, we publish the pilot's per-instance values (Table 7) so that the reversal is auditable rather than asserted: the pilot means are Base 0.500 against BundleS 0.333—i.e. -16.7pp, BundleS *worse*—against the prospective panel's $+12.5$pp with BundleS better. The sign of the descriptive contrast flips. The ordering of the third condition flips too: the pilot places TypedContract level with BundleS and below Base, whereas the prospective panel places it highest. Both batches used identical conditions, grader and two repetitions per instance-condition cell, so the reversal is not attributable to a protocol change.

Statistical procedure (frozen; descriptive sensitivity only). The task instance (n=12) is the independent unit; two repetitions are nested observations (72 trajectories are *not* treated as n=72). Pilot instances 0001–0003 are excluded from the primary contrast. Each per-instance, per-condition rate is the mean over its two repetitions. The primary contrast is the signed per-instance difference $d_i = \mathrm{BundleS}_i - \mathrm{Base}_i$. *Exact paired sign test.* Of the 12 instance differences, 5 are non-zero (3 positive, 2 negative; 7 ties), so the test is conditioned on n=5 non-zero differences with k=3 positive;

Table 7: Three-instance STA pilot: per-instance correct-rates over two repetitions, published so the direction reversal against the prospective panel (Table 6) can be checked. Generated from the frozen collection report by the same committed script that produces the interval estimates.

instance	Base	BundleS	TypedContract	BundleS−Base
0001	0.5	0.0	0.0	−0.5
0002	1.0	1.0	1.0	0.0
0003	0.0	0.0	0.0	0.0
mean	**0.500**	**0.333**	**0.333**	**−0.167**

the two-sided exact p-value is $2\sum_{i=0}^{\min(k,n-k)}\binom{n}{i}/2^n = 1.0$, excluding ties (standard convention). This is a non-establishment result, not evidence of a zero effect. *Per-instance permutation sensitivity.* Under the sharp null of no condition effect, we reshuffle the six within-instance condition labels (2 Base, 2 BundleS, 2 TypedContract) independently per instance, recompute $\sum_i(\mathrm{BundleS}_i - \mathrm{Base}_i)$, and report the two-sided proportion of 10^4 Monte-Carlo replicates (fixed seed 20260811) whose absolute re-randomized sum meets or exceeds the observed $\sum_i d_i = 1.5$, giving $p = 0.31$. This is a descriptive sensitivity check over the fixed instance set, *not* a population-rate p-value (the instances are not sampled from a defined population). No adaptive k was used; no trajectory-pooled p-value is reported; results are reported at the preregistered analysis.

Post hoc sensitivity (added after the frozen analysis; not preregistered). The frozen procedure above yields a test but no effect size with any indication of stability, and a p-value of 1.0 invites exactly the reading the data do not support—that the effect is zero. We therefore add a percentile bootstrap that resamples the 12 instances *with replacement* (the declared independent unit; 10^5 replicates, fixed seed), giving +12.5pp, and we report the *central 95% of that instance-resampling distribution*, which runs from −12.5 to +41.7; we call it the 95% sensitivity band. **We do not call it a confidence interval.** The declared estimand is the realized twelve-instance panel, and a realized panel's composition has no sampling distribution; resampling its members therefore does not quantify uncertainty about that estimand, it perturbs the panel. Because the draws are with replacement, each replicate duplicates some instances and omits others, so what is perturbed is the panel's empirical weighting: the band is a panel-composition sensitivity, and the 95% names a quantile range of that perturbation distribution, not a coverage probability. It is not used for hypothesis testing, is not used to change the preregistered decision rule, and licenses no claim beyond the panel.

What the band does not cover. It quantifies sensitivity to panel composition only. It does not quantify trajectory-level Monte-Carlo uncertainty: with two repetitions per instance–condition cell, each cell's latent success probability is itself poorly resolved, and that component of variability appears nowhere in the band. We deliberately do not add a third interval or a hierarchical model to absorb it—that would blur the preregistered/post-hoc boundary this appendix exists to keep sharp—and state it instead as a limit on what the band may be read to mean. Every number is generated from the frozen records by a committed script and re-checked against the frozen report's own declared values.

D WHAT WOULD CHANGE EACH CONCLUSION

§7 states that we declare in advance what evidence would move each verdict; none of it requires trusting our interpretation. These are those conditions.

- *S0–S1 (observed and replicated here) would be weakened or revised* by a preregistered counterbalanced replication on the same frozen instances that fails to reproduce the Base–BundleS separation, or by a demonstration that the bundle leaks the answer—that a solver could recover the unique assignment from BundleS without consulting the evidence. C6 is separable and the frozen condition files are released so this is checkable; we flag it as the single most informative experiment not yet run.
- *S2-M (not established) would be resolved either way* by raising k on DeepSeek-V4-Pro under the same counterbalancing. Our 3/4 = 3/4 cannot distinguish an absent effect from one too small for k=4; we claim only that it is unestablished.

- *S2-F (not established on STA) could be resolved toward transfer* by a prospectively frozen panel powered to distinguish a preregistered, practically meaningful effect from zero. We do not treat S0–S1 as implying a cross-family effect size, so such a panel's target must be preregistered on its own grounds.
- *S3 (unmeasured) can only be populated, not resolved by re-analysis:* it needs a second backend crossed with an independent family. No combination of the projections fills it—even had S2-M and S2-F both succeeded, no configuration in S3 would have been run—so nothing here licenses a prediction about it.
- *The ceiling result (SPICE) would become informative* under a harder generator for the same family: a ceiling is a property of our instances, not of the family.
- *The mechanism claim (none isolated) would be revised* by any single component reproducing the axis-suppression pattern across two independently collected run windows—a repeated-measurement test of stability, not a scope replication (§3.1). The two C24 rows of Table 5 are exactly this test, and they disagree.
- *The audit protocol's coverage claim would be strengthened or refuted* by applying the frozen rubric to a second external repair corpus; our single probe found the sampling layer entirely unsupported.

E Claim-qualification standard — full tier definitions

§3.1 states the four tiers compactly; their full definitions, including the distinctions that decide borderline cases, are these.

- **Observed.** A descriptive separation in the measured sample at that support. Carries no uncertainty claim and no target beyond E.
- **Replicated.** The direction reappears, under the same protocol, at *newly frozen configurations that enlarge the projection being claimed*: a new instance replicates in s_I, a second backend in s_M, an independent family in s_F. Re-running the *same* configurations in a later window is *repeated measurement*—evidence about stability, not about scope—and we never count it as replication. Because instances are family-specific, a probe that enlarges s_F necessarily enlarges s_I too, so a cross-family replication is never a replication in s_F alone.
- **Estimated on a fixed panel.** A point estimate over a declared, prospectively frozen panel, reported with an explicit account of how far it moves under resampling of that panel's members. Inference is to the realized panel and no further ($T = E$); because a realized panel's composition has no sampling distribution, what accompanies the estimate is a *sensitivity band*, not a confidence statement about a population.
- **Generalized.** Inference to a defined population of instances, models or families—the one tier asserting $T \supsetneq E$—which requires a sampling or generator distribution the design actually instantiates.

F Semantic/tool discrimination — exclusion accounting and pairing sensitivity

This appendix supports §3.1. It is a retrospective, post-freeze derived analysis: no episode, model call or tool run, every value re-derived from records committed at or before the experiment freeze by a committed script with a `--check` mode that recomputes and diffs against its output.

stratum	n	sem. correct	sem. wrong	tool accepts	wrong *and* accepted
workflow (program-primary)	49	33	16	49	16/16
STA, prospective panel	72	24	48	72	48/48
STA, phase-5C	12	5	7	12	7/7
STA, phase-5D	18	7	11	18	11/11
SPICE, phase-5D	18	18	0	18	— (negative control)
retained total	169	87	82	169	82/82

The retained total is shown for completeness only; it is *not* a population rate and the strata are never pooled for any reported claim.

The two verdicts. The semantic verdict is the typed oracle's: for the workflow family the submitted tuple compared to the hidden golden tuple by the same frozen classifier the Study I ledger uses; for STA and SPICE the frozen `semantic_binding` field (submitted tuple = golden). The tool-success signal is each family's own pre-existing field—`tool_exit` together with `signoff` from the stage-2 evidence record (workflow), `pt_signoff_green` (STA), `simulation_success` (SPICE). Nothing is introduced for this analysis: a stratum recording no tool-success field is excluded, never imputed. The STA grader already computed the conjunction of interest as a marker (`green_but_unattested`) without ever aggregating it.

Exclusions: 33 of 202, never imputed. 12 controlled-pair episodes survive only as per-cell counts, with no per-episode record and therefore no tool field. 12 SPICE phase-5C episodes carry no graded tool component. 9 workflow episodes fail verdict-to-artifact pairing. We report the excluded counts rather than a bare retained total so the retained set cannot be mistaken for the whole program.

Pairing, and why it is decided by hash. A tool verdict may be compared with a semantic verdict only if both describe the same final submission. For STA and SPICE this holds by construction: the hidden runner produces the tool verdict at grading time from the submitted binding, leaving no interval in which the submission can drift. The workflow family is different—the agent runs its two-stage evidence chain during the episode and may then edit `flow_config.json`—so pairing is verified by comparing the configuration hash the stage-2 record declares it consumed against the hash of the preserved submission. Nine of 58 fail. Comparing tuples instead of hashes detects only two of the nine: the other seven agree on (scenario, corner, netlist) while the consumed file differs, and one submits a different netlist entirely. The frozen grader flags these independently through its stage-chain component, so the strict rule is primary precisely so that the derived statistic agrees with the grader rather than contradicting it.

The conclusion does not depend on the pairing rule. Under the strict hash rule the workflow stratum retains 49 episodes with 16 semantic errors; under tuple agreement, 56 with 22; with pairing not enforced, 58 with 24. In all three the tool-success signal is constant at *accept*, so every semantic error is accepted: 16/16, 22/22, 24/24. The looser rules are reported only as this sensitivity check and are not used anywhere else.

What this does and does not license. The families are deliberately constructed so a wrong binding remains tool-green, so a false-accept proportion of 1.0 demonstrates the construction operating as specified and quantifies its cost to measurement. It is not an estimate of false-accept prevalence in agent benchmarks generally, and the strata—spanning stages, conditions, models and run windows—are not one sampling frame and are never pooled into a single rate. Because the tool signal is constant, $\Delta \equiv 1 - S_{\text{semantic}}$ and is not an independent quantity. Finally, that the oracle rejects every wrong binding whenever the constraints admit exactly one satisfying assignment is a construction property of the generator, not a theorem.

G Study III — full 26-task Terminal-Bench repair coding

Rubric frozen before coding; source: PR #53 "Terminal-Bench 2.1" (body sha256 `c7172fc6`; files sha256 `fccf2d61`); 2.0/2.1 snapshots frozen (89 tasks each; sha256 `5c7ed05c`).

Task	threat	mechanism	coverage
adaptive-rejection-sampler	Cap	instruction clarification	partial
build-pmars	Cap	instruction clarification; verifier/test tightening	partial
caffe-cifar-10	Exec; Cap	timeout; verifier/test broadening; instruction; env/dep; resource	partial
compile-compcert	Exec	environment/dependency repair	partial

configure-git-webserver	Cap	oracle repair; verifier/test tightening	partial
extract-moves-from-video	Exec	external-dependency removal	partial
filter-js-from-html	Cap; Exec	instruction clarification; resource increase	partial
fix-git	Art	hidden-artifact removal; verifier/test tightening	direct
hf-model-inference	Cap	verifier/test broadening	partial
install-windows-3.11	Cap	instruction clarification; verifier/test tightening	partial
make-doom-for-mips	Exec	oracle repair; env/dep repair	partial
mteb-leaderboard	Exec; Cap	env/dep; instruction; resource	partial
mteb-retrieve	Exec; Cap	env/dep; instruction clarification	partial
polyglot-c-py	Cap	verifier/test broadening	partial
polyglot-rust-c	Cap	verifier/test broadening	partial
protein-assembly	Cap	oracle repair	partial
rstan-to-pystan	Cap	oracle repair	partial
sam-cell-seg	Cap	instruction clarification	partial
torch-tensor-parallelism	Cap; Exec	instruction clarification; resource increase	partial
torch-pipeline-parallelism	Exec	resource increase	partial
query-optimize	Cap	instruction clarification	partial
train-fasttext	Exec; Cap	verifier/test tightening; env/dep repair	partial
build-pov-ray	Other	env/dep repair	not-covered
financial-document-processor	Other	env/dep repair	not-covered
mcmc-sampling-stan	Other	env/dep repair	not-covered
overfull-hbox	Other	env/dep repair	not-covered

Aggregate: direct 1 / partial 21 / not-covered 4; Cap 17, Exec 10, Art 1, Other 4; Samp 0; 6 multi-layer.

H Operational independence, freeze points, and phase chronology

Family independence is defined by five pre-registered structural criteria (independent templates, vocabularies, truths, graders, decoys).

Which results are exploratory and which are confirmatory. An exploratory result may not be read as a confirmation of the hypothesis it generated, so we state where each freeze fell rather than leaving it to be inferred from the narrative order.

- *Exploratory (hypothesis-generating).* Discovery of the tool-green axis-binding failure; the transport repair that made the measurement valid at all; the workflow controlled pair; and the construction of the clarity bundle. BundleS was **selected** during this stage, so its development-instance cell (S0) is exploratory by construction.
- **Freeze point 1 — intervention and held-out instance.** BundleS was fixed as C1+C2+C4+C7 with C6 withheld, and the held-out instance was frozen, *before* the held-out evaluation ran. The S1 cell is therefore a within-family confirmation of a pre-committed intervention on a pre-committed instance, which is why it is the only cell we describe as replicating anything.
- *Exploratory again.* The cross-model arm and the whole component decomposition (C1; Schema vs. Contract; C2-only, C4-only, C24 and the C24 bridge) are exploratory localization; we label them so in the main text and draw no confirmatory conclusion from them.

- **Freeze point 2 — cross-family study.** For the prospective STA study the instance panel, its size (n=12), the conditions, the randomization schedule, and the analysis (exact paired sign test, with the permutation sensitivity as a declared secondary) were all committed *before any paid call of that phase*. Sample size was not adapted and no analysis was selected after seeing outcomes. The instance-resampling sensitivity band of Appendix C was added afterwards and is labelled as post hoc wherever it appears.
- *Confirmatory.* The prospective STA panel is this study's only preregistered confirmatory test of transfer, and it did not establish an effect.

TypedContract entered as a *pre-specified secondary* contrast in the cross-family collection; no primary hypothesis about it was preregistered, which is why its descriptively highest rate is reported as hypothesis-generating and nothing further. The three-instance pilot preceded the prospective panel and is never pooled with it.

Program accounting. Episodes 58/24/36/72; costs ¥682.25 / ¥7.72 / ¥11.75 / ¥43.56. The 58 is the workflow program-primary accounting declared by the frozen ablation-phase manifests; the 70-episode descriptive Study I ledger of Table 5 is broader because it additionally includes the 12 earlier controlled-pair episodes, which predate that accounting and its cost ledger.

I INFRASTRUCTURE + CUSTODY

Exact-commit worktree; canonical-hash guard (`FAILED_INTEGRITY` stop); episode arbiter; SSE streaming; tool-health sentinel; immutable-core/derived-deck repair; per-episode custody byte-matching. Pre-flight caught a grading-path bug (malformed signoff JSON) before the prospective run; historical pilot unaffected (0/30 triggers).

J POSITIONING RELATIVE TO PRIOR WORK

Table 9: Positioning relative to related work. The right column states what this paper adds beyond each line of prior work; the Related Work section gives the same positioning in prose for the works it cites, and the paragraph below covers the three closest concurrent works and the transfer-dimension literature.

Prior work	What it establishes	What this paper additionally does
Harness-Bench (Yao et al., 2026)	Broad model×harness characterization	Claim-scope lattice + controlled semantic mechanism
Rethinking (Wang et al., 2026a)	Leaderboard conflation; search/overfitting	Scope coordinates are partially ordered, not nested
HarnessOpt-Bench (Ursekar et al., 2026)	Optimizer capability	Studies validity of harness-effect claims
ABC (Zhu et al., 2025) / Protocol validity (Shao et al., 2026)	Setup; shortcut/reward validity	Per-control threat log
AFTER (Belikova et al., 2026)	Local/cross-task/cross-role/cross-model transfer of procedural skills	Claim indexed by its configuration support; unmeasured separated from unestablished

The three closest concurrent works, and the transfer-dimension literature. Liu et al. (2026) audits trajectory-level safety properties of harness execution (boundary compliance, execution fidelity, system stability; 210 tasks across 8 domains); our object differs—we audit the evidentiary scope and measurement validity of semantic-binding performance claims, not execution-trajectory safety. Raj et al. (2026) localize a failure to a model/harness/tool/environment interaction edge—*where* a failure originates, where our layers ask whether the *measurement* of it was valid. Caban (2026) argue from psychometric validity that agentic evaluation compounds reliability failures, the concern our sampling and execution layers instantiate empirically. None indexes the evidence for a harness-effect claim by the support it occupies. Reporting transfer along several named dimensions is not itself new either: Belikova et al. (2026) evaluate procedural-memory skills under separate local, cross-task, cross-role and cross-model settings, finding that some transfer broadly while others specialize to a role. Our contribution is not that observation but the indexing—locating the *evidence* for a claim by the set of configurations it occupies, which separates a hypothesis measured and left unestablished from one never measured at all, together with a qualification standard tied to that index.

Ethics statement. The preregistered blinded human construct-validity study (Study B) is preregistered but unexecuted because qualified independent human annotators were unavailable; no LLM annotator was substituted.

AI-use statement (outside the page limit). Generative-AI assistants were used to support experimental design and critique, orchestration and infrastructure code, task generators, analysis and report scripts, literature search, result-interpretation checks, and manuscript drafting and editing. All experimental decisions, task definitions, acceptance criteria, statistical analyses, and scientific claims were reviewed and approved by the authors. Experimental results were produced by frozen executable pipelines and independently checkable graders reproduced from frozen manifests; generative-AI systems were not used as ground-truth judges or scientific oracles. The authors take responsibility for all content and conclusions.

Reproducibility statement. Deterministic generators (fixed seeds); independent per-family graders; exact-commit isolated-worktree execution with pre/post-episode canonical-hash verification; committed sample-membership arbitration; validity-only replacement; per-episode custody byte-matching. Program totals: 58+24+36+72 paid primary episodes; committed-ledger cost ¥745.29. (The descriptive Study I ledger of Table 5 reports 70 workflow episodes: these 58 plus the 12 earlier controlled-pair episodes, which sit outside this program accounting and its cost ledger.) Frozen manifests, randomization schedules, custody hashes, per-episode sanitized evidence, and the scripts that re-derive every table from the preserved records are in the anonymized supplement. ICLR 2027 deadlines: abstract September 18, 2026 AOE; full paper September 25, 2026 AOE.